

Space Shuttle Orbiter



START OF A MISSION

The Space Shuttle Columbia is slowly rolled out of its hangar to the launch pad where it will await the countdown. A typical Shuttle mission usually lasted about a week – this one was a service mission to the Hubble Space Telescope.

THE SHUTTLE WAS A SPACE VEHICLE shaped by compromises enforced by budgetary constraints. NASA chose the basic configuration in 1972 as above all affordable to develop. The orbiter would be a delta-winged craft capable of gliding to land after reentering the atmosphere. For takeoff there would be an external jettisonable liquid-propellant tank to fuel the main engines and two solid-fuel

“It’s an exciting and dynamic moment. The whole vehicle starts to shake and rattle so much you can barely read the instruments.”

CREW MEMBER JAMES VOSS

REFLECTIONS ON THE LAUNCH AT LIFTOFF

boosters strapped on to the tank to increase thrust. This somewhat ungainly arrangement slashed development costs, but produced a vehicle far removed from dreams of an aircraft that would be flown on wings into space and back.

As well as the three main engines, which proved a formidable technical challenge, the Shuttle was designed with two additional manoeuvring engines and 44 thrusters. To protect it from temperatures that would reach 3,000°F (1,659°C) on reentering the atmosphere at around less than 27,000kph (17,000mph), the orbiter’s skin was covered with around 30,000 silica tiles.

Control thrusters situated in nose (and at rear of Shuttle) are used to change position in space

Two-storey cabin, with upper flight deck and lower mid-deck crew quarters

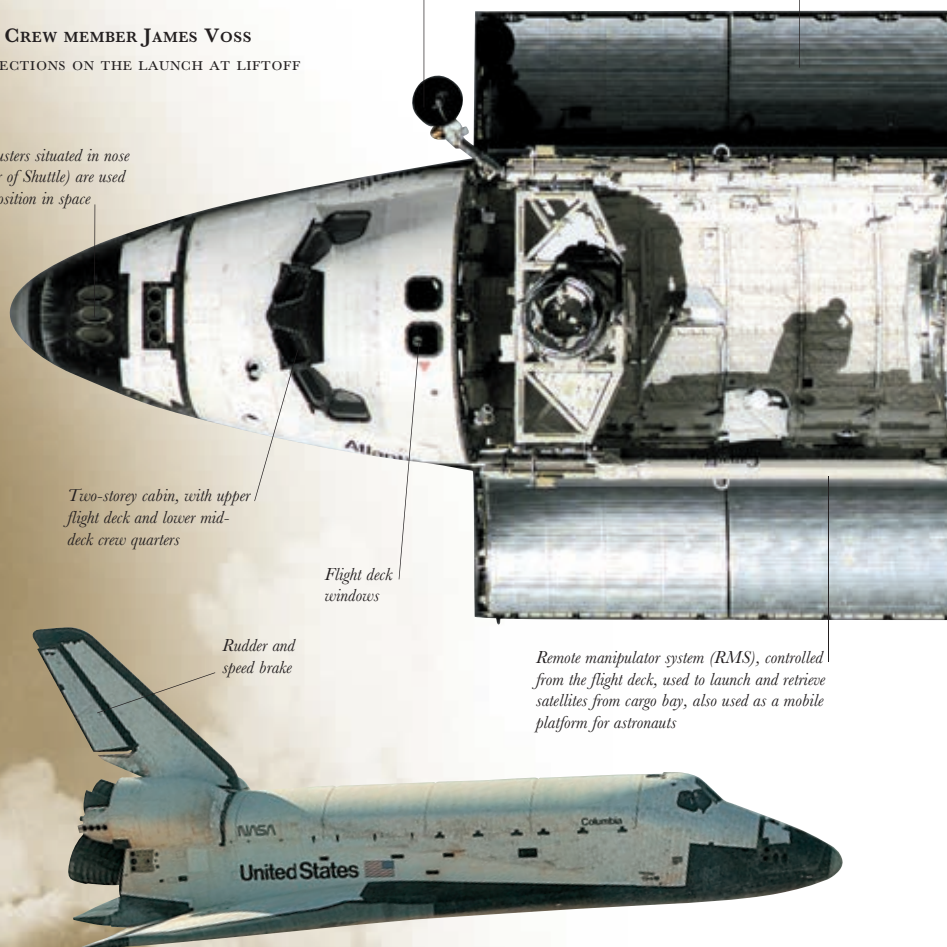
Flight deck windows

Rudder and speed brake

Ku-band antenna

Radiator panels attached to cargo bay doors release heat from Shuttle and stop it from overheating

Remote manipulator system (RMS), controlled from the flight deck, used to launch and retrieve satellites from cargo bay, also used as a mobile platform for astronauts



COLUMBIA MAKES FIRST SHUTTLE FLIGHT

On April 12, 1981, the Space Shuttle Columbia made its maiden flight, lasting just over 54 hours. At the end of its twenty-eighth mission, on February 1, 2003, Columbia disintegrated during reentry with the loss of its seven-person crew.